

Name: _____

Find the image of the point after a reflection in the given line.

1. $A(-3, 5)$ in the x -axis
2. $A(-3, 5)$ in the y -axis
3. $B(6, -2)$ in the line $y = x$
4. $B(6, -2)$ in the line $y = -x$

Use the coordinate planes at the right.

5. Graph $\triangle RST$ with $R(1, 4)$, $S(4, 5)$, and $T(5, 2)$ and its image after a reflection in the line $x = 2$. List the image coordinates.
6. Graph $\triangle DEF$ with $D(-7, 2)$, $E(-4, 4)$, and $F(-3, 1)$ and its image after the glide reflection below. List the final coordinates. Translation: $(x, y) \rightarrow (x + 9, y)$
Reflection: in the x -axis

Answer each question.

7. Find the coordinates of the image of $U\bar{V}$ with $U(-5, 3)$ and $V(-2, 6)$ after a reflection in the line $y = 2$.
8. How many lines of symmetry does a regular octagon have? Explain how you know.
9. How many lines of symmetry does each figure at the right have? (isosceles triangle, parallelogram, square)
10. A classmate says that reflecting a point in the x -axis and then in the y -axis gives the same image as reflecting it in the line $y = x$. Is the classmate correct? Give a counterexample.

